MOBAN Chapter 9 — Recent Evolutions

Study Guide

Outline

---

1. 6G Visions

1.1 The Landscape

Multiple competing 6G visions exist (ITU, SNS IA, HEXA-X, etc.). There is **no single agreed-upon 6G standard yet** — the name refers to a collection of visions targeting the 2030s.

Key themes across all visions:

- Instantaneous interactions and trustworthy control
- Integration of communication, sensing, computation, and AI
- Coverage beyond current 5G, into new frequency ranges
- New service category proposed: **HRLLC** (Hyper-Reliable Low-Latency Communications)

1.2 6G Spectrum

FR3 — Centimeter waves (7–15 GHz): The main candidate band for 6G.

> **Why centimeters, not mmWave?**
>
> - mmWave 5G deployments have largely **failed for mobile traffic**
> - In the US: average time connected to mmWave 5G is **< 1%** of total connected time
> - South Korea cancelled 28 GHz spectrum allocations due to operator "lack of spending"
> - mmWave is viable only for **Fixed Wireless Access (FWA)** — not general mobile use

Band	Frequencies	Role
Sub-6 GHz	< 6 GHz	Wide coverage (same as 5G)
FR3 / Centimeter wave	7–15 GHz	6G primary mobile band
sub-THz	90–300 GHz	Massive data, very short range (niche)

1.3 The Spectrum Battle

Key tension: **exclusive licensed spectrum vs. shared/unlicensed**

- Coverage is limited by **spectrum/regulation**, not technology
- Private licensed spectrum is **technology-neutral** — any standard could use it
- 5G is encroaching into unlicensed Wi-Fi bands (e.g., 6 GHz)
- Question: should we decouple spectrum from technology? Run Wi-Fi in licensed bands?

---

2. Deterministic Connected Systems

2.1 What Is Determinism?

A **deterministic** communication system offers **bounded, guaranteed** end-to-end latency and reliability — not just "best effort" or statistical guarantees.

Current commercial 5G and Wi-Fi do **NOT** offer true deterministic guarantees (they are eMBB-focused).

2.2 Use Cases Requiring Determinism

Domain	Latency Target	Example
Industrial automation	20 μs – 10 ms	Machine-to-machine control
Automotive	Real-time	Safety-critical in-vehicle systems
Audio/video	Low latency	Professional multimedia production
Holographic communications	< 20 ms + Gbps	Immersive interactions
Tactile Internet	< 5 ms	Human-machine haptic interaction
Mission-critical	< 10–100 ms	Emergency/defense operations
Collaborative robotics	10–100 ms	Remote decision-making

2.3 Candidate Technologies for Deterministic Communication

Technology	Spectrum	Use
Wired Ethernet (TSN)	—	Industrial LAN backbone
Wi-Fi (specialized)	Unlicensed / local licensed	80–90% of indoor data
5G (specialized)	Global/local licensed	Outdoor + some indoor

> **Key insight:** 80% of data is consumed **indoors**. 80–90% of all data traffic goes over Wi-Fi (unlicensed). Cellular 5G carries only 10–20% of total traffic. This makes **Wi-Fi the primary target** for deterministic wireless.

2.4 The Problem with COTS Chips

COTS (Commercial Off-The-Shelf) chips are:

- Black-box designs with limited control

- Optimized for mass-market (eMBB) use cases
- Not reconfigurable for specialized deterministic needs

→ Specialized and customizable hardware platforms are needed (e.g., FPGA-based SDR).

---

3. Time-Sensitive Networking (TSN)

3.1 What is TSN?

TSN is a set of **IEEE 802.1 standards** extending Ethernet to provide deterministic, time-synchronized, bounded-latency communication.

Goal: **End-to-end reliable and latency-bounded connectivity**

Key standards:

Standard	Purpose
IEEE 802.1AS	Time synchronization (precision clock)
IEEE 802.1Qbv	Time-aware scheduling (gating mechanism)

3.2 Wired TSN → Wireless TSN

The challenge: extend TSN guarantees from wired Ethernet **over wireless links** (Wi-Fi or 5G).

5G-TSN Integration (5G URLLC)

Architecture: 5G system acts as a **logical TSN bridge** between two TSN segments.

- DS TT (Device Side TSN Translator) at the UE
- NW TT (Network Side TSN Translator) at the UPF
- CNC (Central Network Controller) manages the TSN schedule

> **Critical limitation:** The current 5G-TSN architecture is **NOT truly deterministic**!
>
> - Target: μs timing offset, < 1 ms latency
> - Reality: 5G was designed for eMBB, not TSN. URLLC features (preemption, mini-slots, retransmissions) are layered onto a "massively complex design that does not scale down well."

Wireless TSN over Wi-Fi (UGent IDLab approach)

The **openwifi** platform: a customizable Wi-Fi implemented on FPGA (System-on-Chip).

- Full white-box design: from digital baseband to Linux driver
- High-speed on-chip bus: ping RTT of **300 μs** (vs. 1 ms in COTS Wi-Fi)
- Enables full low-level customization not possible with COTS chips

Key achievements:

Feature	Performance
Time synchronization	µs-level accuracy
IEEE 802.1Qbv gating	Implemented over Wi-Fi
Bounded E2E latency	**< 3.25 ms** (COTS Wi-Fi: ~70 ms)
Seamless mobility (handover)	< 9 ms with UWB + ML support
Impactless association	Beacon-based, no traffic disruption
In-band telemetry (INT)	Real-time E2E latency, RSSI, retransmissions

Three patented techniques:

1. Device and Method for clock synchronization in a wireless network
2. Impactless association in a wireless time-sensitive network
3. Impactless hand-over in a wireless time-sensitive network

3.3 Deterministic Systems Vision (Summary)

> *"Enabling communication stacks and connectivity services with deterministic end-to-end guarantees using programmable wireless & wireline hardware — offering guarantees beyond speed & capacity in complex, dynamic, and mobile environments."*

Philosophy:

- Move away from **one-size-fits-all** (mass-market) solutions
- Target high-end, smaller-volume **professional markets**
- Use a subset of standardized + pre-standard features
- **Prove it** — no theoretical claims without hardware demonstration

---

4. Mission-Critical Communications

4.1 BOLSTER Project (B5G Tactical Network)

Goal: Beyond-5G network for private mobile standalone tactical (military/defense) use.

Key features:

- **Zero-touch** network deployment and management
- Standalone B5G mounted in a vehicle
- Extended coverage via **mobile base station** (urban and rural)
- Ruggedized UEs + surveillance drones

Technology stack:

- Citymesh vehicle with Athonet 5G core + Nokia 5G RAN
- IMEC SDR platform for:
    - **AI/ML-powered technology recognition & traffic characterization**
    - Intelligent Radio Resource Management (RAN slicing, interference mitigation)

- PHY-layer encryption via **CSI obfuscation**

4.2 AI/ML Technology Recognition

Semi-supervised learning for identifying which radio technology is transmitting (Wi-Fi, 5G, LTE, interference, unknown).

- Patent: EP4026059A1 — Neural Network for Identifying Radio Technologies
- Used to detect interference and steer traffic to the right radio resource

4.3 O-RAN Integration

O-RAN (Open RAN): open, disaggregated base station architecture.

Key components:

- **O-CU** (Open Central Unit), **O-DU** (Open Distributed Unit), **O-RU** (Open Radio Unit)
- **Near-RT RIC** (Real-Time RAN Intelligent Controller): < 10 ms control loop
- **Non-RT RIC** / SMO: > 1 s control loop (policy, ML model management)
- **A1 interface** (policy), **E2 interface** (control)

Use case: Intelligent scheduler uses TRTC (Technology Recognition & Traffic Characterization) to detect interference and prioritize high-priority slices in real time.

4.4 5G/6G + Wi-Fi Integration

Goal: Intelligent, real-time integration of 3GPP (5G/6G) and IEEE 802.11 (Wi-Fi/TSN) under the O-RAN framework.

Provides:

- Optimal wireless technology selection and traffic steering
- Real-time monitoring, control, and optimization
- Rapid automated deployment in complex environments
- Improved privacy and security

---

5. Wi-Fi — What's Next?

5.1 Wi-Fi Context

- > 21 billion Wi-Fi devices in use today

- Wi-Fi carries **> 50% of all internet traffic**
- Present in virtually every indoor environment
- No indoor environment without Wi-Fi

5.2 Wi-Fi 7 (IEEE 802.11be) — Extremely High Throughput (EHT)

Wi-Fi 7 is finalized and being deployed now.

Key new features:

Feature	Description
Multi-Link Operation (MLO)	Combine 2+ Wi-Fi bands (2.4 / 5 / 6 GHz) into one logical link. Increases throughput and reliability. Modes: non-simultaneous or simultaneous Tx/Rx
Preamble Puncturing	Transmit on wide channels (80/160 MHz) even if part of the channel is occupied by another AP. Punctures busy 20 MHz sub-channels
Restricted TWT (R-TWT)	Reserved contention-free time intervals negotiated between AP and client. Other devices cannot transmit during R-TWT windows → determinism for latency-sensitive flows
4096-QAM	Higher modulation order → more bits per symbol
320 MHz channels	In 6 GHz band
Multi-RU (MRU)	Flexible resource unit allocation in OFDMA

R-TWT mechanism:

```
Time →
AP:     [BEACON] ... [negotiate R-TWT] ... [trigger] [data] [trigger] [data]
Client:                                               [data]          [data]
Others: ──────────────────────── NO TX ALLOWED ──────────────────────────────────
```

5.3 Wi-Fi 8 (IEEE 802.11bn) — Ultra-High Reliability (UHR)

Wi-Fi 8 is under standardization (target: late 2020s). Focus: **reliability** for high-criticality use cases.

Key new features:

Feature	Purpose
Multi-AP Coordination	Multiple APs work together, coordinated via trigger frames (< 50 ns sync accuracy over optical backhaul)
Coordinated Spatial Reuse (C-SR)	APs coordinate simultaneous transmissions to different STAs, reducing interference. Up to **50% goodput increase**
Coordinated Beamforming	APs steer beams jointly to avoid inter-BSS interference
Coordinated TDMA	APs share time slots to eliminate collision
Coordinated R-TWT	Coordinated contention-free windows across multiple APs
Distributed MLO	Handle roaming by switching between AP links seamlessly (< 9 ms with ML support)
Seamless Roaming	No packet loss during AP handover

Feature	Purpose
Enhanced Long Range PPDU	Extended range transmissions
Non-Primary Channel Access (NPCA)	Use secondary channels opportunistically
Distributed Resource Units (dRU)	Flexible OFDMA across multiple APs

Multi-AP Coordination architecture:

```
AP 1 ──── Coordination (backhaul) ──── AP 2
  |                                       |
STA 1                                 STA 2
       ← C-OFDMA trigger from AP 1 →
              (simultaneous UL)
```

5.4 Industry Standardization Timeline

```
Wi-Fi 6 (802.11ax) → Wi-Fi 7 (802.11be) → Wi-Fi 8 (802.11bn) → ...
    2019                  2024                  ~2028
 OFDMA, TWT           MLO, R-TWT           Multi-AP, C-SR
```

---

6. Joint Sensing & Communications

6.1 What Is ISAC?

ISAC (Integrated Sensing And Communications): Using the same radio hardware and signals for **both communication and sensing/radar** simultaneously.

This is a key 6G capability — the radio network becomes a sensor network.

6.2 RF Sensing Approaches

Approach	Method	Use
Passive spectrum sensing	Analyze IQ samples from SDR	Classify known tech; detect unknown/anomalies
Active radar sensing	Use Wi-Fi or 5G signals as radar	Detect humans, objects, movement
AI-powered analysis	Foundation models, transformers, U-NET, CNNs	Higher accuracy, generalization

RF situational awareness: Combine passive + active sensing for a complete picture of the radio environment.

6.3 Wi-Fi Radar

The communication signal itself can be used as a radar:

- **Monostatic SIMO Wi-Fi Radar**
- AoA (Angle of Arrival) resolution: **~6°**
- Distance change detection: **~1.37 mm**
- Can track the location and movement of other Wi-Fi devices

6.4 Secure Sensing — CSI Obfuscation

Problem: Standard communication signals can be **misused** for sensing (e.g., someone tracking your location through walls using your Wi-Fi).

Solution: CSI Fuzzer

- Extension of the communication system
- Adds controlled noise/perturbation to CSI (Channel State Information)
- **Only authorized parties** can perform sensing → unauthorized sensing is prevented

6.5 AI Wireless Foundation Models

A **foundation model** is a large model pre-trained on massive unlabeled data that can be fine-tuned for many downstream tasks (analogous to GPT/BERT for language).

Applied to wireless: a single model trained on diverse RF data could handle:

- Technology recognition
- Channel estimation
- Anomaly detection
- Beam management
- ... all without task-specific training from scratch

Key Takeaways

Topic	Core Message
6G	No single vision; centimeter waves (7–15 GHz) as primary band; mmWave has failed for mobile
Determinism	Current 5G/Wi-Fi are NOT deterministic; specialized hardware (FPGA/SDR) is needed
TSN	IEEE 802.1AS (sync) + 802.1Qbv (gating); 5G-TSN exists but is not truly deterministic; Wi-Fi TSN with < 3.25 ms achievable
Mission-critical	O-RAN + AI/ML for intelligent, adaptive, self-managing tactical networks

Topic	Core Message
Wi-Fi 7	MLO + R-TWT + preamble puncturing → higher throughput and first steps toward determinism
Wi-Fi 8	Multi-AP coordination (C-SR, C-TDMA) → ultra-high reliability for critical applications
Joint sensing	Same radio = communication + radar; CSI obfuscation for privacy; AI foundation models for RF

Study guide created for MOBAN Chapter 9: Recent Evolutions — UGent IDLab (Ingrid Moerman, Jeroen Hoebeke)